



F6 Logarithmic graphs

Name: _____

Class: _____

Date: _____

Time: **99 minutes**

Marks: **83 marks**

Comments:

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Q1.

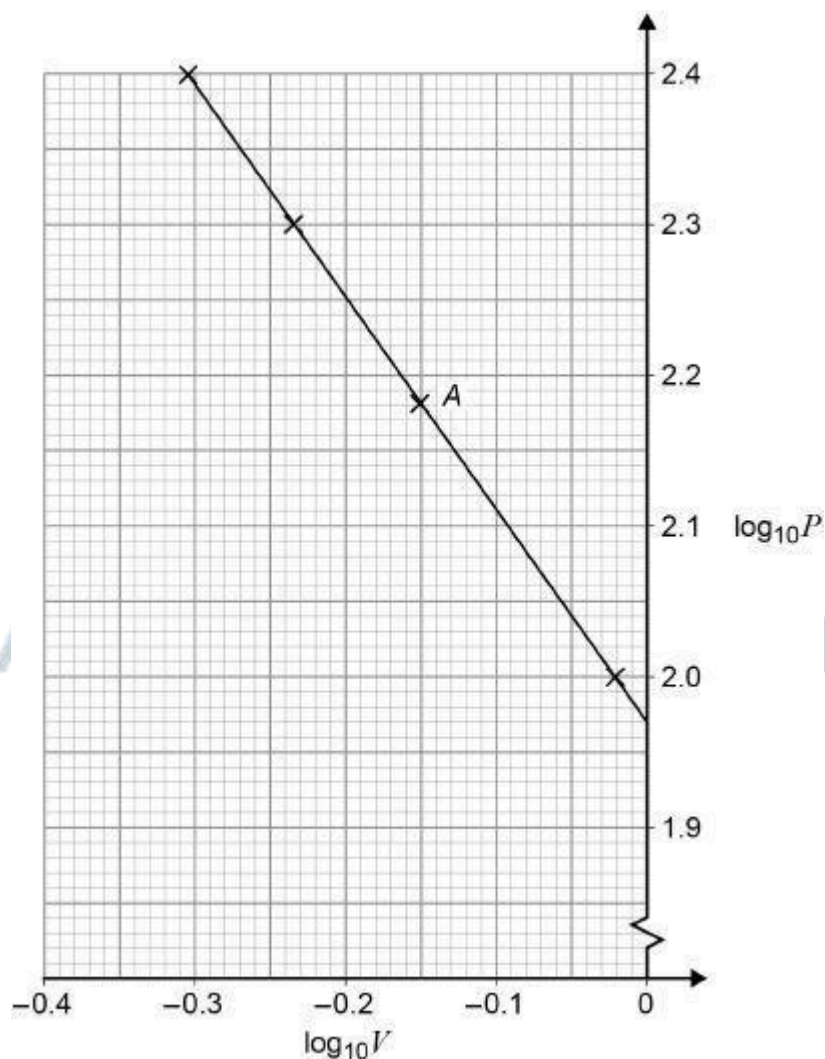
Maxine measures the pressure, P kilopascals, and the volume, V litres, in a fixed quantity of gas.

Maxine believes that the pressure and volume are connected by the equation

$$P = cV^d$$

where c and d are constants.

Using four experimental results, Maxine plots $\log_{10} P$ against $\log_{10} V$, as shown in the graph below.



(a) Find the value of P and the value of V for the data point labelled A on the graph.

(2)

(b) Calculate the value of each of the constants c and d .

(4)

- (c) Estimate the pressure of the gas when the volume is 2 litres.

(2)

(Total 8 marks)

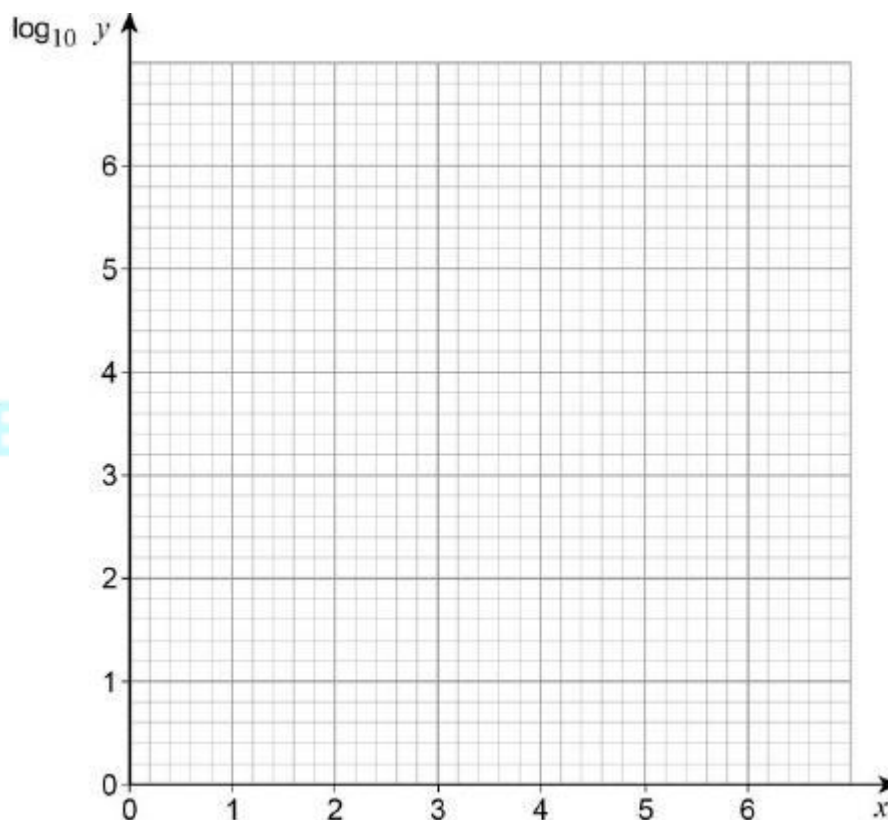
Q2.

A student conducts an experiment and records the following data for two variables, x and y .

x	1	2	3	4	5	6
y	14	45	130	1100	1300	3400
$\log_{10} y$						

The student is told that the relationship between x and y can be modelled by an equation of the form $y = kb^x$

- (a) Plot values of $\log_{10} y$ against x on the grid below.



(2)

- (b) State, with a reason, which value of y is likely to have been recorded incorrectly.

(1)

- (c) By drawing an appropriate straight line, find the values of k and b .

(4)

(Total 7 marks)

Q3.

The variables y and x are related by an equation of the form

$$y = ax^n$$

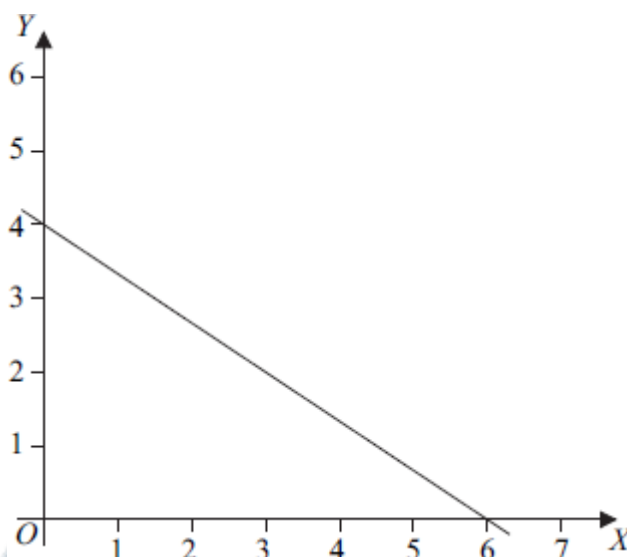
where a and n are constants.

Let $Y = \log_{10} y$ and $X = \log_{10} x$.

- (a) Show that there is a linear relationship between Y and X .

(3)

- (b) The graph of Y against X is shown in the diagram.



Find the value of n and the value of a .

(4)

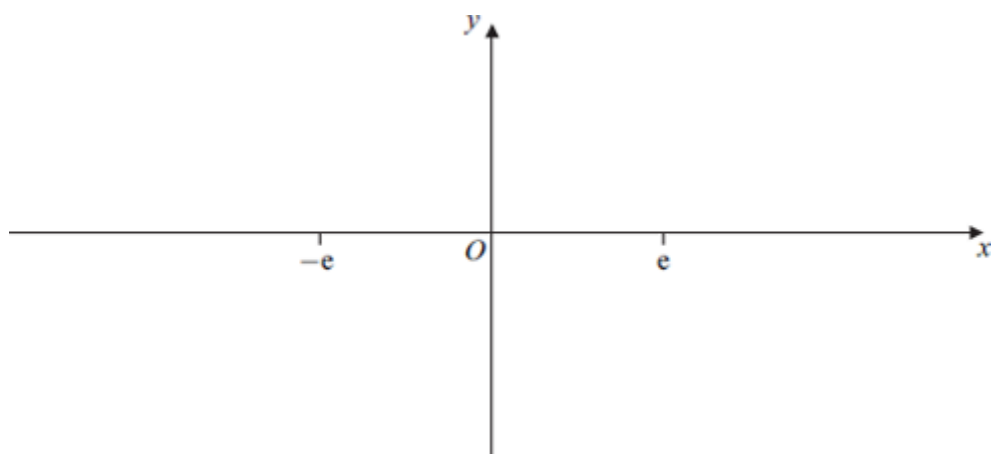
(Total 7 marks)

Q4.

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x - e)$.

(4)

- (b) Sketch, on the axes given below, the graph of $y = |4 \ln(x - e)|$, indicating the exact value of the x -coordinate where the curve meets the x -axis.



(3)

- (c) (i) Solve the equation $|4 \ln(x - e)| = 4$.

(3)

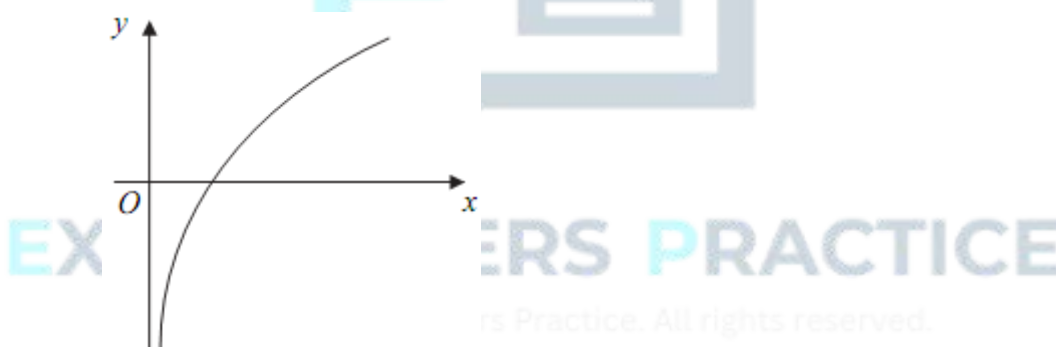
- (ii) Hence, or otherwise, solve the inequality $|4 \ln(x - e)| \geq 4$.

(3)

(Total 13 marks)

Q5.

The diagram shows the curve with equation $y = \ln(6x)$.



- (a) State the x -coordinate of the point of intersection of the curve with the x -axis.

(1)

- (b) Find $\frac{dy}{dx}$.

(2)

(Total 3 marks)

Q6.

The variables x and Y , where $Y = \log_{10} y$, are related by the equation

$$Y = mx + c$$

where m and c are constants.

(a) Given that $y = ab^x$, express a in terms of c , and b in terms of m .

(3)

(b) It is given that $y = 12$ when $x = 1$ and that $y = 27$ when $x = 5$.

On the diagram below, draw a linear graph relating x and Y .

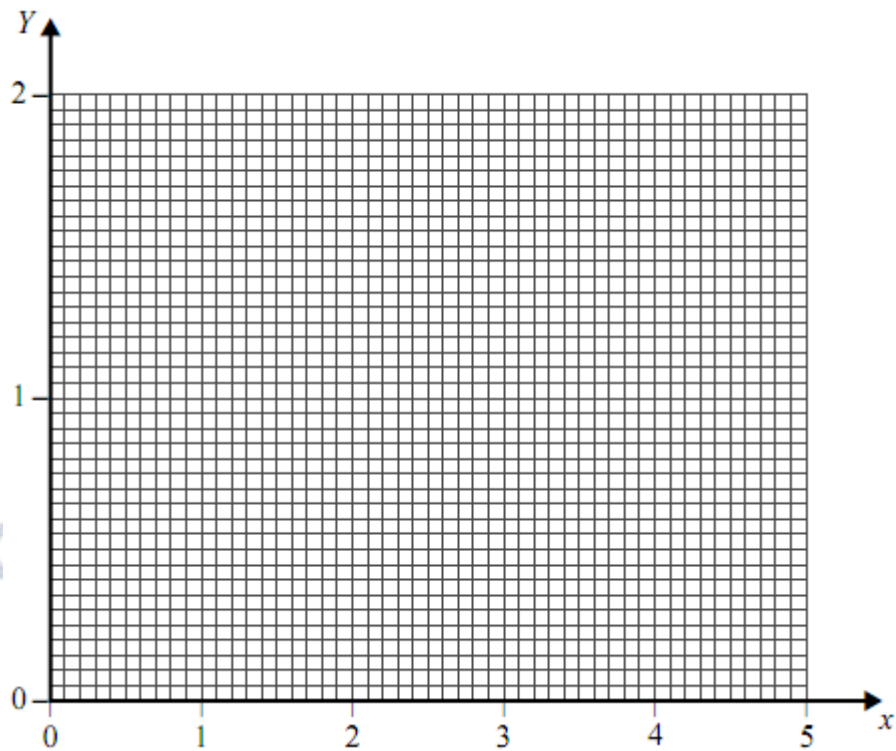
(3)

(c) Use your graph to estimate, to two significant figures:

(i) the value of y when $x = 3$;

(2)

(ii) the value of a .



(2)

(Total 10 marks)

Q7.

The variables x and y are known to be related by an equation of the form

$$y = ab^x$$

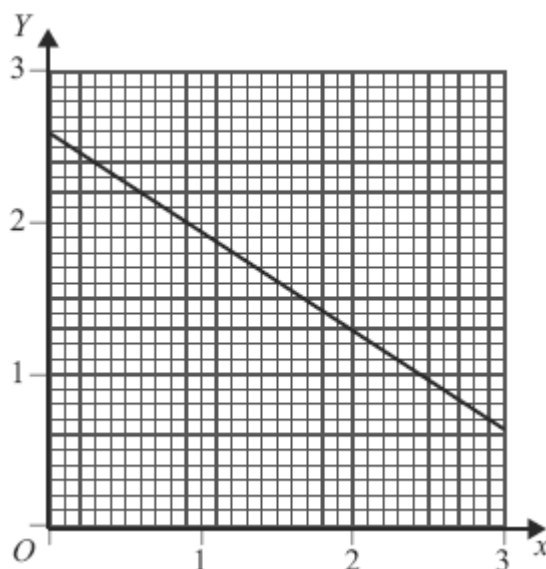
where a and b are constants.

(a) Given that $Y = \log_{10} y$, show that x and Y must satisfy an equation of the form

$$Y = mx + c$$

(3)

- (b) The diagram shows the linear graph which has equation $Y = mx + c$.



Use this graph to calculate:

- (i) an approximate value of y when $x = 2.3$, giving your answer to one decimal place;
- (ii) an approximate value of x when $y = 0.8$, giving your answer to one decimal place.

(You are not required to find the values of m and c .)

(4)

(Total 7 marks)

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Q8.

The variables x and y are related by an equation of the form

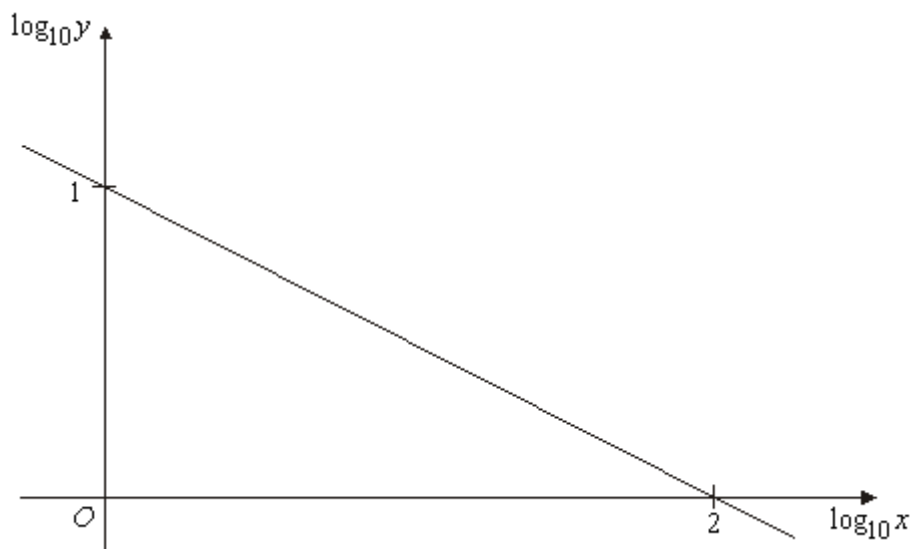
$$y = ax^b$$

where a and b are constants.

- (a) Using logarithms to base 10, reduce the relation $y = ax^b$ to a linear law connecting $\log_{10}x$ and $\log_{10}y$.

(2)

- (b) The diagram shows the linear graph that results from plotting $\log_{10}y$ against $\log_{10}x$.



Find the values of a and b .

(4)

(Total 6 marks)

Q9.

The variables x and y are known to be related by an equation of the form

$$y = ab^x$$

where a and b are constants.

The following approximate values of x and y have been found.

x	1	2	3	4
y	3.84	6.14	9.82	15.7

- (a) Complete the table below, showing values of x and Y , where $Y = \log_{10} y$.
Give each value of Y to three decimal places.

x	1	2	3	4
Y	0.584			

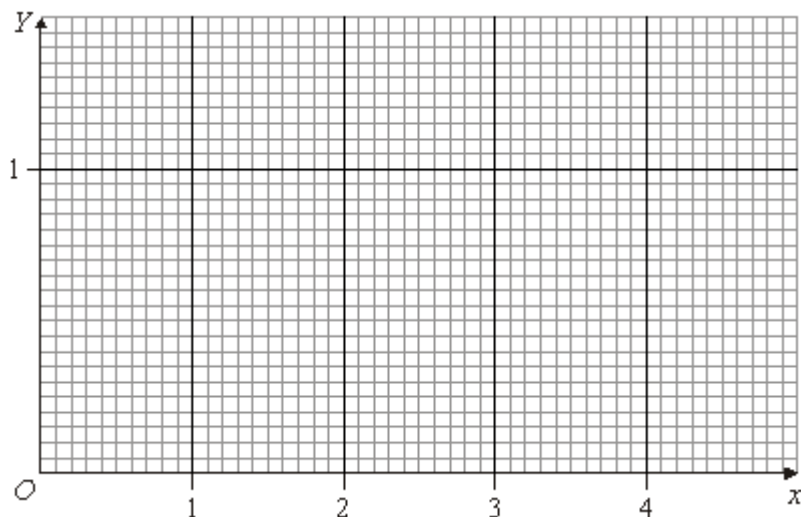
(2)

- (b) Show that, if $y = ab^x$, then x and Y must satisfy an equation of the form

$$Y = mx + c$$

(3)

- (c) Draw on the diagram below a linear graph relating x and Y .



(2)

- (d) Hence find estimates for the values of a and b .

(4)

(Total 11 marks)

Q10.

The variables x and y are known to be related by an equation of the form

$$y = kx^n$$

where k and n are constants.

Experimental evidence has provided the following approximate values:

x	4	17	150	300
y	1.8	5.0	30	50

- (a) Complete the table in **Figure 1**, showing values of X and Y , where

$$X = \log_{10} x \quad \text{and} \quad Y = \log_{10} y$$

Give each value to two decimal places.

X	0.60			2.48
Y	0.26			1.70

Figure 1

(3)

- (b) Show that if $y = kx^n$, then X and Y must satisfy an equation of the form

$$Y = aX + b$$

(3)

- (c) Draw on **Figure 2** a linear graph relating X and Y .

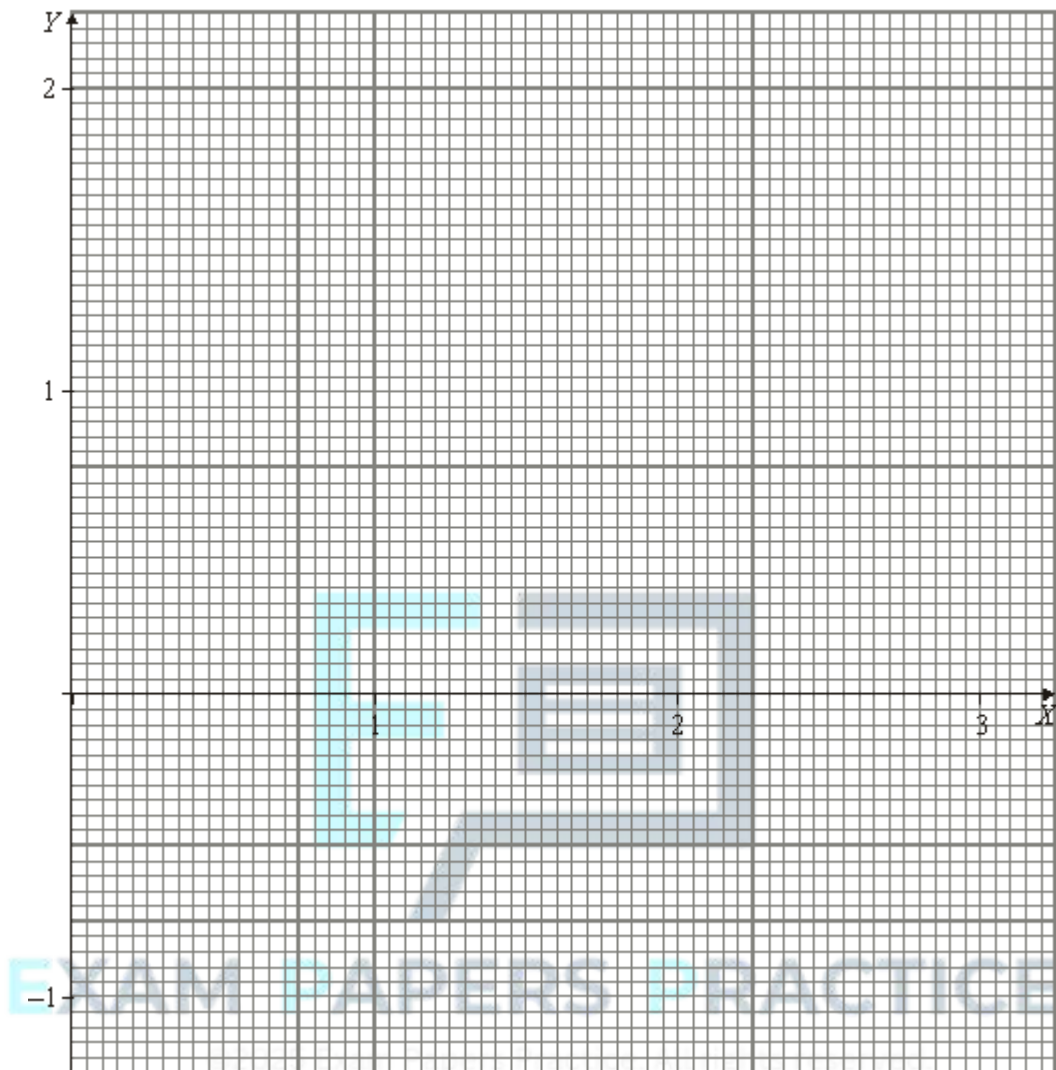


Figure 2

(3)

- (d) Find an estimate for the value of n .

(2)

(Total 11 marks)